

Answer Key: Algebra Foundations Shadow Skills Check-up

1. Collecting Like Terms

- a) $-9a + 9b$
- b) $-5x^2y - 6xy + 4xy^2$

2. Multiply with Powers

- a) $12p^5q^5$
- b) $12a^4b^6c^8$

3. Divide with Powers

- $\frac{x^3}{3y^4}$ (or $\frac{1}{3}x^3y^{-4}$)

4. Powers of Powers

- $64a^6b^{15}$

5. Substitution

- $W = \frac{3(15) - 5(-4)}{5} = \frac{45 + 20}{5} = \frac{65}{5} = 13$

6. Rearrange Formula

- $x = \frac{A}{5p}$

7. Expanding Brackets

- Single: $12x^2y - 15y^4$
- Quadratics 1: $x^2 + 3x - 28$
- Quadratics 2: $-4x^2 + 4x + 15$

8. Factorising

- Single: $6ab(4a - 2b + 1)$
- Quadratics 1: $(x + 7)(x + 3)$
- Quadratics 2: $(x - 7)(x + 4)$

9. Special Case Factorisation

- Perfect Square: $(x - 6)^2$ or $(x - 6)(x - 6)$
- Difference of Two Squares: $(x - 7)(x + 7)$

10. Non-Unit Coefficient Factorisation

- $2(x^2 + 7x + 10) = 2(x + 5)(x + 2)$

- $(3x - 2)(x + 3)$

11. Algebraic Fractions

- Multiplication: $\frac{4(x-2)}{3} \times \frac{x+2}{8} = \frac{x^2-4}{6}$
- Addition: $\frac{3(x-3)+4(x+5)}{12} = \frac{3x-9+4x+20}{12} = \frac{7x+11}{12}$

12. Solving Linear Equations

- One/Two Step: $x = 5$
- One/Two Step: $x - 10 = 24 \Rightarrow x = 34$
- x on both sides: $8x = 32 \Rightarrow x = 4$
- Fractional: $(x + 3)(x - 8) = (x - 2)(x - 5)$
 $x^2 - 5x - 24 = x^2 - 7x + 10$
 $2x = 34 \Rightarrow x = 17$

13. Linear Inequalities

- $-3x + 6 < -9$ $-3x < -15$ $x > 5$ (Note: Sign flips when dividing by a negative)

14. Simultaneous Equations

- Elimination: Multiply top by 5, bottom by 3: $20x + 15y = 90$ $6x - 15y = 66$
 $26x = 156 \Rightarrow x = 6$ $4(6) + 3y = 18 \Rightarrow 3y = -6 \Rightarrow y = -2$ Result: $x = 6, y = -2$
- Substitution: $5x - 2(12 - 3x) = 31$ $5x - 24 + 6x = 31$ $11x = 55 \Rightarrow x = 5$
 $y = 12 - 3(5) \Rightarrow y = -3$ Result: $x = 5, y = -3$

End of Answer Key